Task 4 : Voice Translator

Introduction

Voice Translator focuses on the development of a system that converts spoken English audio into Hindi text using machine learning techniques. The system integrates speech recognition, a custom-trained neural machine translation model, and a graphical user interface to enable real-time interaction. The project emphasizes practical implementation of artificial intelligence concepts rather than relying solely on pre-built translation services.

The Voice Translator is designed with specific operational constraints, including language restriction and time-based availability. These constraints help simulate real-world system behavior where access control and system availability are important considerations. This report documents the overall development process, learning experience, challenges faced, and outcomes achieved during the execution of the project.

Background

Speech-based language translation has become increasingly important in modern communication systems, particularly in multilingual environments. Most existing translation applications rely on large pre-trained models, offering limited insight into how translations are generated. This project takes a different approach by focusing on training a custom English–Hindi translation model using deep learning techniques.

The project is based on neural machine translation using sequence-to-sequence architecture with an attention mechanism. By integrating speech recognition and a graphical user interface, the system provides both technical depth and usability. The inclusion of time-based access control further enhances the system by introducing practical operational constraints.

Learning Objectives

One of the primary objectives of this project was to gain hands-on experience in building and training a neural machine translation model from scratch. This included understanding data preprocessing, tokenization, padding, and training strategies required for sequence-based models.

Another important objective was to integrate multiple components into a single functional system. By combining speech recognition, machine learning inference, and GUI development, the project aimed to strengthen system integration skills and improve understanding of real-world AI application development.

Activities and Tasks

The project was carried out through a structured sequence of activities beginning with dataset preparation. The English–Hindi dataset was cleaned by removing duplicates and null values, followed by tokenization and padding to prepare the data for training. A sequence-to-sequence LSTM model with attention was then designed and trained using TensorFlow and Keras.

After training the model, speech recognition functionality was implemented to capture English audio input from the user. A graphical user interface was developed using Tkinter to display recognized English text and translated Hindi output. All components were integrated and tested to ensure proper functionality within the specified time window.

Data :

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Samples: 11071
```

	english	hindi
0	I have to go to sleep.	<start> मुझे सोना है। <end>
1	Muiriel is 20 now.	<start> म्यूरियल अब बीस साल की हो गई है। <end>
2	Muiriel is 20 now.	<start> म्यूरियल अब बीस साल की है। <end>
3	The password is "Muiriel".	<start> कूटशब्द "Muriel" है। <end>
4	The password is "Muiriel".	<start> पासवर्ड "Muriel" है। <end>
5	I will be back soon.	<start> मैं जल्द लौटूंगी। <end>
6	I'm at a loss for words.	<start> मैं तो लाजवाब हो गयी हूँ। <end>
7	This is never going to end.	<start> यह तो कभी खत्म न होगा। <end>
8	I just don't know what to say.	<start> मुझे नहीं पता मैं क्या कहूँ। <end>
9	That was an evil bunny.	<start> वह ख़रगोश दुष्ट था। <end>
10	Education in this world disappoints me.	<start> मैं इस दुनिया में शिक्षा पर बहुत निराश...
11	That won't happen.	<start> वैसा नहीं होगा। <end>
12	I miss you.	<start> मुझें तुम्हारी याद आ रही है। <end>
13	I miss you.	<start> मुझें आपकी याद आ रही है। <end>
14	You should sleep.	<start> तुम्हें सोना चाहिए। <end>
15	You should sleep.	<start> आपको सोना चाहिए। <end>
16	I never liked biology.	<start> मुझे जीव विज्ञान कभी भी पसंद नहीं था। ...
17	No I'm not; you are!	<start> मैं नहीं हूँ, तुम हो! <end>
18	That's MY line!	<start> वह तो मेरी लाईन है! <end>
19	Are you sure?	<start> पक्का? <end>

Model :

```
Model: "seq2seq_model"
__________________________________________________________________________________________________
 Layer (type)                   Output Shape         Param #     Connected to
==================================================================================================
 encoder_inputs (InputLayer)    [(None, 53)]         0           []

 decoder_inputs (InputLayer)    [(None, 59)]         0           []

 encoder_embedding (Embedding)  (None, 53, 300)      2226900     ['encoder_inputs[0][0]']

 decoder_embedding (Embedding)  (None, 59, 300)      2306700     ['decoder_inputs[0][0]']

 encoder_lstm (LSTM)            [(None, 53, 256),    570368      ['encoder_embedding[0][0]']
                                 (None, 256),
                                 (None, 256)]

 decoder_lstm (LSTM)            [(None, 59, 256),    570368      ['decoder_embedding[0][0]',
                                 (None, 256),                     'encoder_lstm[0][1]',
                                 (None, 256)]                     'encoder_lstm[0][2]']

 attention_layer (Attention)    (None, 59, 256)      1           ['decoder_lstm[0][0]',
                                                                  'encoder_lstm[0][0]']

 concatenate (Concatenate)      (None, 59, 512)      0           ['decoder_lstm[0][0]',
                                                                  'attention_layer[0][0]']
...
Total params: 9,618,794
Trainable params: 9,618,794
Non-trainable params: 0
__________________________________________________________________________________________________
```

Testing :

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Statement is : Don't cause a commotion.
Prediction is : हल्ला मत मचाओ।

Statement is : Is this Tom's?
Prediction is : टॉम का है?

Statement is : It's in your hands.
Prediction is : आपके हाथों में है।

Statement is : I want money.
Prediction is : मुझे पैसा चाहिए।

Statement is : This is a great place.
Prediction is : यह बढ़िया जगह है।

Statement is : If you help me learn English, I'll help you learn Japanese.
Prediction is : अगर तुम ने मेरी अंग्रेज़ी सीखने में मदद सीखने में मदद करूँगी।

Statement is : You have a lot to learn.
Prediction is : आपको कुछ ज़्यादा ही तेज़ है।

Statement is : It was easy.
Prediction is : आसान था।
...
Prediction is : मैं यूनान जाऊँगी।

Statement is : I meant it as a joke.
Prediction is : मैंने तो मज़ाक समझकर बोला था।
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Skills and Competencies

This project significantly enhanced technical skills related to machine learning and natural language processing. Working with TensorFlow and Keras improved understanding of neural network training, while implementing attention-based translation models strengthened knowledge of advanced NLP concepts.

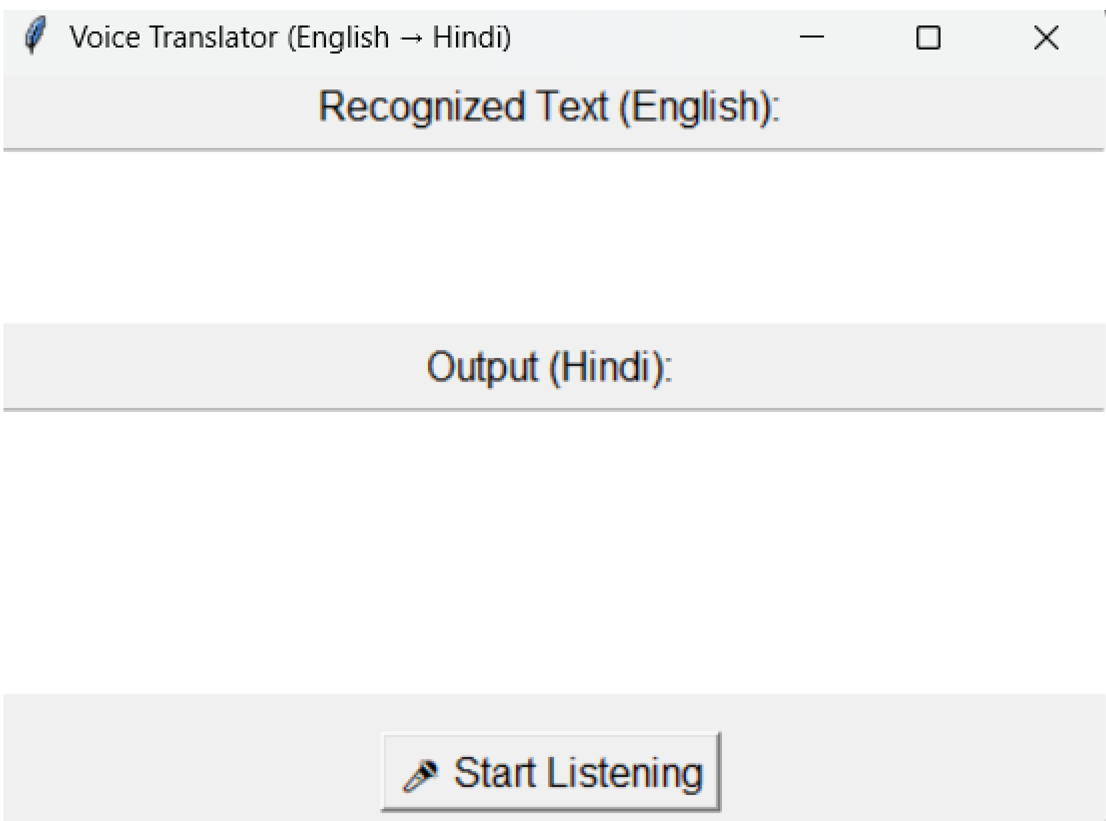
In addition to technical skills, the project also developed competencies such as debugging, logical reasoning, and system integration. Designing a GUI and handling user interaction improved software development skills, while writing this report strengthened technical documentation and communication abilities.

Feedback and Evidence

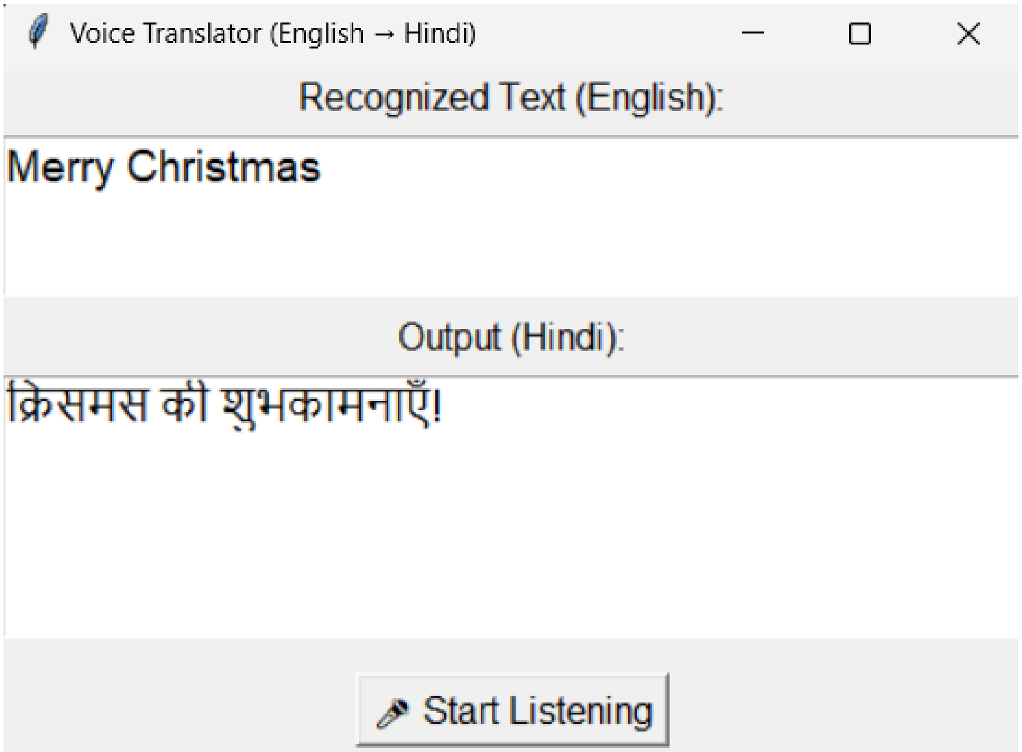
Throughout the development process, regular testing provided feedback on the system's performance and translation accuracy. Different English speech inputs were tested to evaluate recognition quality and translation output, allowing continuous improvement of the system.

Evidence of successful implementation includes trained machine learning model files, tokenizer files, GUI execution results, and displayed translations. Screenshots of model training, recognized speech, and translated output serve as proof of system functionality and learning outcomes.

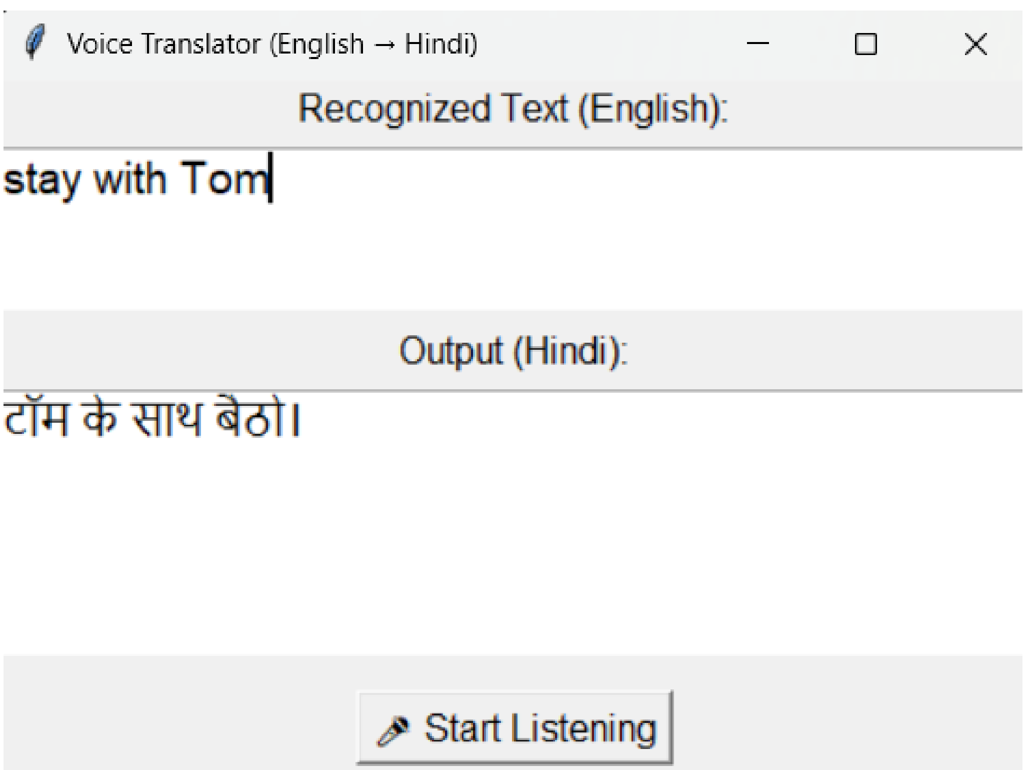
GUI :



GUI Working :

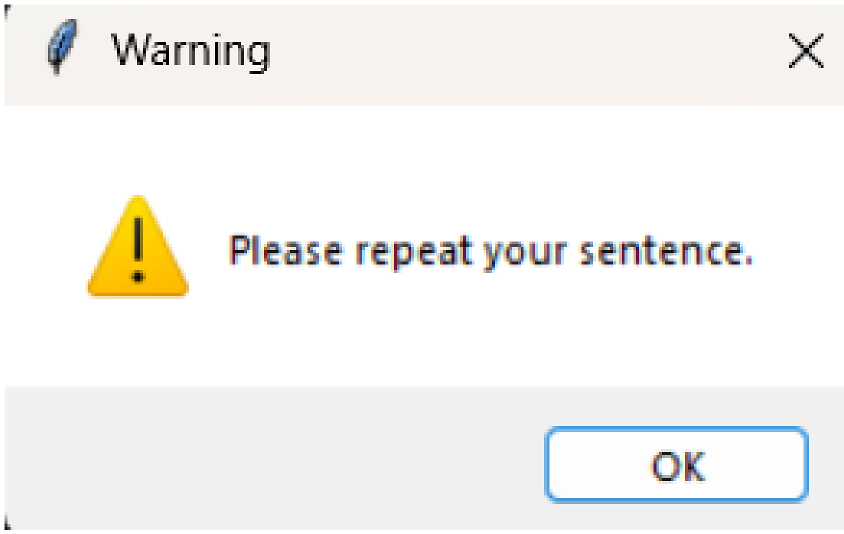


Voice Translator (English → Hindi)
Recognized Text (English):
Merry Christmas
Output (Hindi):
क्रिसमस की शुभकामनाएँ!
Start Listening



Voice Translator (English → Hindi)
Recognized Text (English):
stay with Tom
Output (Hindi):
टॉम के साथ बैठो।
Start Listening

GUI Error :



Challenges and Solutions

One of the major challenges faced during the project was handling unclear or noisy speech input, which sometimes resulted in recognition errors. This issue was addressed by implementing exception handling that prompts the user to repeat the input when the system fails to understand the audio.

Another challenge was maintaining translation accuracy for sentences of varying length. This was resolved by using an attention mechanism in the sequence-to-sequence model. Implementing time-based access control was also challenging and was successfully handled using system time validation logic.

Outcomes and Impact

The final outcome of the project is a functional Voice Translator system capable of converting English speech into Hindi text within a defined time window. The system demonstrates successful integration of speech recognition, deep learning, and GUI-based interaction.

The impact of this project extends beyond technical implementation, as it provided valuable experience in building real-world AI systems. The knowledge gained will be useful for future academic projects and professional work in artificial intelligence and machine learning.

Conclusion

The Voice Translator project successfully achieved all its intended objectives and demonstrated the effective application of deep learning and natural language processing techniques. The system provides accurate translation, user-friendly interaction, and controlled access through time-based restrictions.

In conclusion, this project offered meaningful hands-on experience and strengthened understanding of AI system design and deployment. It lays a strong foundation for future enhancements such as multilingual translation, improved speech recognition accuracy, and expanded system capabilities.